

KIRAN RAJEEV

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EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Amrita Vishwa Vidyapeetham • Coimbatore, Tamil Nadu, India • 2027 • CGPA: 8.9

EXPERIENCE

Full-Stack Engineer | Anokha & Pragati (University Event Platform)

- Drove frontend development for the Pragati main website and an integrated multi-event ecosystem, leveraging TypeScript, React, and Tanstack, delivering a performant user experience for Anokha (5,000+ attendees) and Pragati (1,000+ attendees)
- Single-handedly architected and maintained the EventAdmin dashboard utilized by 50+ internal staff, building a robust, reusable interface enabling non-technical staff to rebrand and reconfigure the platform across events without code changes.
- Engineered a web-based attendance app using camera-based QR scanning, handling check-in/out for both individual participants and multi-member teams with live state sync.
- Co-managed deployment of a multi-service architecture across two bare-metal servers; configured Prometheus and Grafana dashboards for real-time system observability.

PROJECTS

paper-scout | Go, Gin, PostgreSQL, Redis, Qdrant, Gemini API, Docker

- Architected a concurrent 7-stage AI data pipeline in Go to automate research analysis, utilizing a bounded worker pool and Redis for state persistence and crash recovery.
- Designed a highly resilient API integration layer featuring a custom circuit breaker and token-bucket rate limiter to reliably manage traffic across Gemini, Semantic Scholar, arXiv, and GROBID.
- Engineered a hybrid Retrieval-Augmented Generation (RAG) system combining vector similarity search (Qdrant) and LLM re-scoring to accurately discover, evaluate, and rank relevant research papers.

VOX | Python, Flask, faster-whisper, OpenAI API, SQLite

- Engineered a real-time voice-to-text desktop pipeline using a single-threaded audio capture path with daemon worker threads and a bounded queue, achieving sub-second transcription latency.
- Integrated Voice Activity Detection (VAD) to filter non-speech audio before processing, significantly reducing hallucinated transcriptions and improving overall Word Error Rate (WER).
- Built an optional LLM post-processing layer using a universal OpenAI-compatible backend (Ollama by default) to clean and reformat raw transcriptions, with a SQLite + FTS5 history store for full-text search across past transcriptions.

Memoria | Go, Bitcask Architecture, LSM-Tree Concepts, Concurrency

- Architected a Bitcask-style key-value storage engine from scratch in pure Go, featuring $O(1)$ point lookups via an in-memory KeyDir and sequential append-only writes.
- Engineered a thread-safe concurrency model using multi-tier read/write locks, safely handling 200+ concurrent operations across reads, writes, and deletes without data races.
- Developed a hint file system with CRC32 integrity checks and an LRU file cache significantly reducing recovery time on restart by avoiding full data-file re-scans.

SKILLS

- **Languages:** Go, TypeScript, JavaScript, Python, C++
- **Frontend:** React, Next.js, TanStack, Tailwind CSS, Zustand, Framer Motion

- **Backend:** Gin, Express.js, Hono JS, Flask, Prisma
- **Data:** PostgreSQL, MySQL, SQLite, Redis, Qdrant, Neon, Supabase
- **Infra:** Docker, Git/Github, Linux, AWS EC2/S3, GCP, Prometheus, Grafana
- **AI:** Gemini API, OpenRouter, Ollama, ONNX Runtime